國立臺北科技大學
National Taipei University of Technology

資料結構

Data Structure

Lab 05

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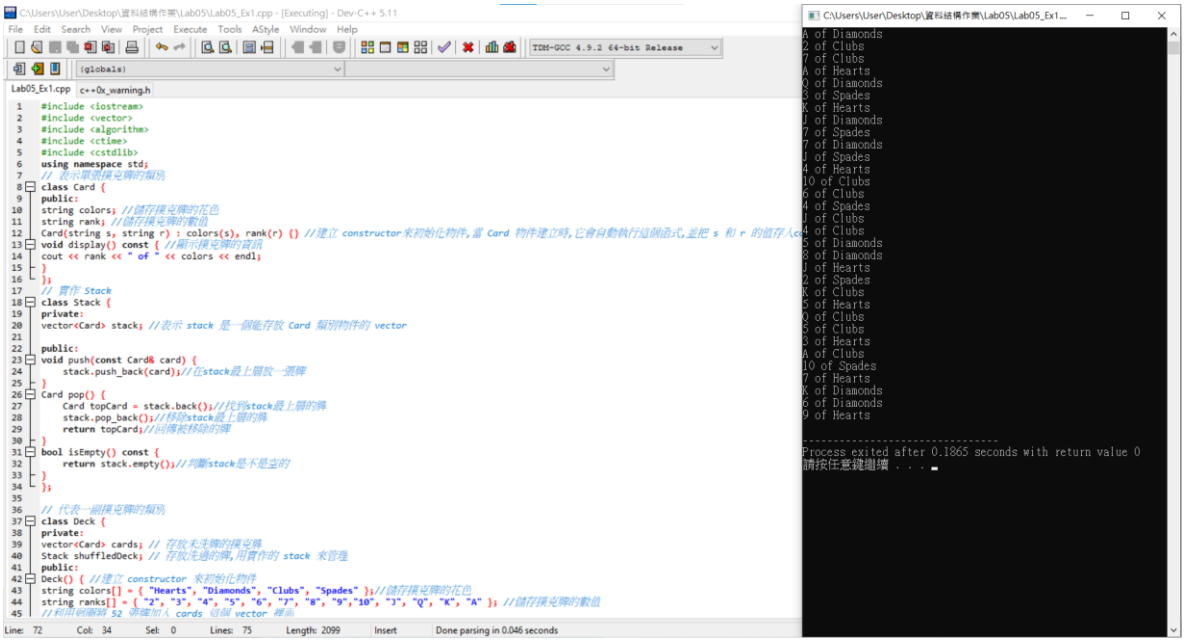
Lab05-Ex1.
用 stack 實作出撲克牌的洗牌與發牌
Code

```
#include <iostream>
#include <vector>
#include <algorithm>
#include <ctime>
#include <cstdlib>
using namespace std;// 表示單張撲克牌的類別
class Card {
public:
string colors; //儲存撲克牌的花色
string rank; //儲存撲克牌的數值
Card(string s, string r) : colors(s), rank(r) {} //建立 constructor 來初始化物件,
當 Card 物件建立時,它會自動執行這個函式,並把 s 和 r 的值存入 colors 和 rank
void display() const { //顯示撲克牌的資訊
cout << rank << " of " << colors << endl;
}
};
// 實作 Stack
class Stack {
private:
vector<Card> stack; //表示 stack 是一個能存放 Card 類別物件的 vector

public:
void push(const Card& card) {
        stack.push_back(card);//在 stack 最上層放一張牌
}
Card pop() {
        Card topCard = stack.back();//找到 stack 最上層的牌
    stack.pop_back();//移除 stack 最上層的牌
    return topCard;//回傳被移除的牌
}
bool isEmpty() const {
        return stack.empty();//判斷 stack 是不是空的
}
};
```

```
// 代表一副撲克牌的類別
class Deck {
private:
vector<Card> cards; // 存放未洗牌的撲克牌
Stack shuffledDeck; // 存放洗過的牌,用實作的 stack 來管理
public:
Deck() { //建立 constructor 來初始化物件
string colors[] = { "Hearts", "Diamonds", "Clubs", "Spades" };//儲存撲克牌的花色
string ranks[] = { "2", "3", "4", "5", "6", "7", "8", "9","10", "J", "Q", "K",
"A" }; //儲存撲克牌的數值
//利用迴圈將 52 張牌加入 cards 這個 vector 裡面
for (int i = 0; i < 4; i++) { // 四種花色
for (int j = 0; j < 13; j++) { // 13 種點數
cards.push_back(Card(colors[i], ranks[j]));
}
}
}
//洗牌
void shuffleDeck() {
    srand(time(0));
    random_shuffle(cards.begin(), cards.end());//洗牌

    for (int i = 0; i < cards.size(); i++) {
        shuffledDeck.push(cards[i]);//放到stack 裡
    }
}
//發牌
void drawAllCards() {
        while (!shuffledDeck.isEmpty()) {
        Card card = shuffledDeck.pop();//取出stack 最上層的牌
        card.display();//顯示取出的牌
    }
}
};
int main() {
Deck deck; //建立 deck 產生 52 張撲克牌
deck.shuffleDeck(); //進行洗牌並放入堆疊
```

```
cout << "Shuffled deck:" << endl;
deck.drawAllCards(); //依序取出堆疊內的牌並顯示
return 0;
}
```

Result



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TDM-GCC 4.9.2 64-bit Release

(globals)

Lab05_Ex1.cpp | c++0x_warning.h

```cpp
#include <iostream>
#include <vector>
#include <algorithm>
#include <ctime>
#include <cstdlib>
using namespace std;
// 表示撲克牌的類別
class Card {
public:
string colors; //儲存撲克牌的花色
string rank; //儲存撲克牌的數值
Card(string s, string r) : colors(s), rank(r) {} //建立 constructor 來初始化物件,當 Card 物件建立時,它會自動執行這個語式,並把 s 和
void display() const { //顯示撲克牌的資訊
cout << rank << " of " << colors << endl;
}
};
// 實作 Stack
class Stack {
private:
vector<Card> stack; //表示 stack 是一個能存放 Card 類別物件的 vector

public:
void push(const Card& card) {
    stack.push_back(card);//在stack最上層放一張牌
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Card pop() {
    Card topCard = stack.back();//找到stack最上層的牌
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    return topCard;//回傳被移除的牌
}
bool isEmpty() const {
    return stack.empty();//判斷stack是不是空的
}
};

// 代表一副撲克牌的類別
class Deck {
private:
vector<Card> cards; // 存放未洗牌的撲克牌
Stack shuffledDeck; // 存放洗過的牌,用實作的 stack 來管理
public:
Deck() { //建立 constructor 來初始化物件
string colors[] = { "Hearts", "Diamonds", "Clubs", "Spades" };//儲存撲克牌的花色
string ranks[] = { "2", "3", "4", "5", "6", "7", "8", "9","10", "J", "Q", "K", "A" }; //儲存撲克牌的數值
//利用迴圈將 52 張牌加入 cards 這個 vector 裡面
```

Line: 72 Col: 34 Sel: 0 Lines: 75 Length: 2099 Insert Done parsing in 0.046 seconds

C:\Users\User\Desktop\資料結構作業\Lab05\Lab05_Ex1.exe

```
Q of Diamonds
6 of Clubs
3 of Hearts
J of Diamonds
5 of Diamonds
3 of Diamonds
J of Clubs
J of Hearts
8 of Hearts
10 of Hearts
9 of Spades
J of Spades
Q of Clubs
6 of Diamonds
3 of Clubs
A of Spades
5 of Hearts
3 of Spades
4 of Hearts
2 of Diamonds
2 of Clubs
9 of Clubs
A of Diamonds
8 of Diamonds
A of Clubs
4 of Diamonds
10 of Spades
9 of Hearts
Q of Spades
7 of Clubs
7 of Spades
K of Clubs
8 of Spades
9 of Diamonds
10 of Diamonds
Q of Hearts

--------------------------------
Process exited after 0.1529 seconds with return value 0
請按任意鍵繼續 . . .
```

C:\Users\User\Desktop\資料結構作業\Lab05\Lab05_Ex1.cpp - [Executing] - Dev-C++ 5.11

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TDM-GCC 4.9.2 64-bit Release

(globals)

Lab05_Ex1.cpp | c++0x_warning.h

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#include <algorithm>
#include <ctime>
#include <cstdlib>
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public:
string colors; //儲存撲克牌的花色
string rank; //儲存撲克牌的數值
Card(string s, string r) : colors(s), rank(r) {} //建立 constructor 來初始化物件,當 Card 物件建立時,它會自動執行這個語式,並把 s 和 r 的值存入
void display() const { //顯示撲克牌的資訊
cout << rank << " of " << colors << endl;
}
};
// 實作 Stack
class Stack {
private:
vector<Card> stack; //表示 stack 是一個能存放 Card 類別物件的 vector

public:
void push(const Card& card) {
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Card pop() {
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}
bool isEmpty() const {
    return stack.empty();//判斷stack是不是空的
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// 代表一副撲克牌的類別
class Deck {
private:
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string ranks[] = { "2", "3", "4", "5", "6", "7", "8", "9","10", "J", "Q", "K", "A" }; //儲存撲克牌的數值
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Line: 72 Col: 34 Sel: 0 Lines: 75 Length: 2099 Insert Done parsing in 0.046 seconds

C:\Users\User\Desktop\資料結構作業\Lab05\Lab05_Ex1.exe

```
J of Hearts
8 of Clubs
4 of Spades
3 of Diamonds
9 of Clubs
Q of Spades
K of Spades
A of Diamonds
9 of Spades
A of Spades
A of Clubs
10 of Diamonds
2 of Clubs
5 of Spades
J of Diamonds
2 of Hearts
7 of Hearts
3 of Spades
2 of Diamonds
5 of Clubs
Q of Hearts
Q of Clubs
8 of Hearts
K of Diamonds
Q of Diamonds
8 of Diamonds
6 of Hearts
4 of Clubs

--------------------------------
Process exited after 0.1771 seconds with return value 0
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```